

Executive Summary

Project overview

This study examined the machine learning approaches for wine classification and wine quality (Red and White) prediction using two datasets, which are Wine Recognition (178 samples) and Wine Quality (Red with 1,599 samples), and (White with 4,898 samples). We implemented systematic hyperparameter tuning to optimize model performance and also evaluated and compared results against published benchmarks.

Methodology

Data Preprocessing

- Standardization of all features to ensure equal contribution.
- Train-test splitting into 80/20.
- Validation using 5-fold cross-validation during model development.

Model Selection and Tuning:

- Wine Recognition (Classification): Logistic Regression and K-Nearest Neighbours (KNNs).
- Wine Quality (Regression): Linear Regression and Multi-Layer Perceptron (MLP).
- Extensive hyperparameter tuning via GridSearchCV and RandomizedSearchCV.

Results

Classification Models:

Model	Best Parameters	CV Accuracy	Test Accuracy
Logistic Regression	C=1, max_iter=1000, penalty=l2, solver=lbfgs	0.9889	1.0000
KNN	algorithm=auto, n_neighbors=11, p=1, weights=uniform	0.9776	1.0000

Both classification models achieved perfect test accuracy which demonstrates the high separability of wine cultivars based on chemical attributes.

Regression Models

Model	Dataset	Best Parameters	CV RMSE	TEST RMSE	TEST R ²
Linear Regression	Red Wine	copy_X=True, fit_intercept=True, positive=False	0.6607	0.6203	0.4113
MLP Regressor	Red Wine	max_iter=1000, learning_rate=adaptive, hidden_layer_sizes=(25,), alpha=0.0001, activation=tanh	0.6603	0.5236	0.5804
Linear Regression	White Wine	copy_X=True, fit_intercept=True, positive=False	0.7622	0.7489	0.2758
MLP Regressor	White Wine	max_iter=1000, learning_rate=constant, hidden_layer_sizes=(25,), alpha=0.0001, activation=relu	0.7429	0.6475	0.4587

MLP Regressors substantially outperformed Linear Regression for both wine quality datasets. Also red wine quality dataset showing better predictability than white wine.

Comparison with Prior Research

Classification models (100% accuracy) surpassed the 78-96% reported by Dhaliwal et al. (2022), likely due to extensive hyperparameter tuning and careful preprocessing.

For wine quality prediction, direct comparison with Ramya et al. (2023) is challenging as they treated wine quality as a classification problem. But some of findings align with their conclusion such as:

- Non-linear models outperform linear ones for wine quality prediction
- Red wine quality is more predictable than white wine
- Even optimal models explain only moderate variance in quality ratings

Conclusion

1. **Classification vs. Regression Performance:**

Classification tasks significantly outperformed regression tasks which suggests limitations in predicting subjective quality ratings from objective chemical measurements.

2. **Model Selection:**

Neural networks (MLP) outperformed simpler linear models for regression which confirms non-linear relationships between chemical properties and wine quality.

3. **Dataset Characteristics:**

Red wine quality showed better predictability than white wine, with 27% higher R^2 values (0.58 vs 0.46).